

Research paper

Depositional constraints and diagenetic pathways controlling petrophysics of Middle Miocene shallow-water carbonate reservoirs (Leitha limestones), Central Paratethys, Austria-Hungary

Ahmed Ali^{a,b,*}, Michael Wagreich^b, Michael Strasser^c

^a Geology Department, Faculty of Science, Minia University, 61519 Menia, Egypt

^b Department of Geodynamics and Sedimentology, University of Vienna, 1090 Vienna, Austria

^c Institute of Geology, University of Innsbruck, 6020 Innsbruck, Austria



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ABSTRACT

Middle Miocene shallow-water limestones, the so-called Leitha limestones, comprise a widespread Middle Miocene lithological unit of the Central Paratethys Sea. It includes mainly limestones composed of corallinean algae and other fossil debris and were deposited in shoals of various basins such as the Vienna, Eisenstadt-Sopron and Styrian basins of Austria where they form potential hydrocarbon reservoir rocks. Petrophysical parameters including porosity, permeability, density, P-wave velocity and magnetic susceptibility measurements were performed and evaluated. The sampled sites of Mannersdorf and Wöllersdorf quarries and drill-hole Bruckneudorf are located in the Vienna Basin and share the same principal limestones depositional facies that is massive, closely-packed rudstone facies. This facies has low porosities below 10%, low permeability lower than 1.2 mD except one samples at 68.5 mD, high bulk density of dry samples up to 2.64 g/cm³ and high P-wave velocity up to 6761 m/s. In Mannersdorf, a second facies type of soft porous coarse-grained rudstone shows significantly higher porosity (up to 47%), higher permeability up to 124.3 mD, lower bulk density of dry samples less than 1.8 g/cm³ and lower P-wave velocity less than 2798 m/s. This latter facies retains primary porosity and secondary porosity enhanced by dissolution. In contrast, the former facies type suffered intensive cementation, which destroyed the primary porosity, and in some cases, subsequent later dissolution and karstification enhanced secondary porosity partly.

Kummer (Austria) and Fertőrákos (Hungary) quarries are located within the Eisenstadt-Sopron Basin. The well-sorted grainstone facies has high porosity up to 50%, extremely high permeability up to 31238 mD, low bulk density of dry samples less than 1.71 g/cm³ and low P-wave velocity less than 3219 m/s. The poorly-sorted massive rudstone facies recorded only in Kummer quarry is less porous (porosities range from 22 to 31%) and less permeable. The role of dissolution predominates in these limestones besides compaction and pressure solution in the massive rudstone facies. Limestones from the Rosenberg quarry of the Styrian Basin show two facies types: (1) wackestone to packstone facies with lower porosity (up to 16.4%) and lower permeability ranges from 0.16 to 0.91 mD, compared to (2) the packstone facies with higher porosity (21%) and higher permeability of 1.83 mD. The influence of diagenesis on both facies is similar, represented by early marine cementation followed by partial dissolution and finally meteoric blocky cementation.

The depositional influence on petrophysics of Leitha limestones, for instance, packing, sorting and grain size represents the basis for the differences in facies types in the same locality and among various localities. This depositional influence controls the subsequent stepwise diagenetic evolution the facies. Local depositional environments and their spatial evolution play a significant role on reservoir properties such as, at Mannersdorf the two facies types are varied due to the primary depositional characteristics and hence, the diagenetic evolution. Diagenetic pathways are principally similar in the different localities and basins; however, different stages of diagenesis are recorded due to different burial and exhumation histories.

* Corresponding author. Geology Department, Faculty of Science, Minia University, 61519 Menia, Egypt.
E-mail address: ahmad.ali@mu.edu.eg (A. Ali).

1. Introduction

Carbonates and sandstones represent the major proportion of hydrocarbon reservoirs worldwide. Carbonate hydrocarbon reservoirs include both dolomites and limestones (Sun, 1995), with a slightly greater percentage of world hydrocarbon reserves have been produced up to now from dolomites as such rocks often, but not always, have more porosity and permeability than limestones (Halley and Schmoker, 1983). Due to the various depositional, diagenetic and tectonic processes, carbonate reservoirs have strongly differing rock fabric elements (Murray, 1960; Dürrast and Siegesmund, 1999; Maliva et al., 2009). Petrophysics of carbonate rocks is strongly affected by two main intrinsic features: (1) carbonates are mostly autochthonously deposited (“grown”) due to organisms, and (2) carbonate minerals are highly reactive (Choquette and Pray, 1970; Moore, 2001). These features build the fundamental differences between carbonates and siliciclastic rocks (Ehrenberg and Nadeau, 2005) and consequently result in significant differences of petrophysical properties and their evolution in time.

Studying petrophysical features (i.e. porosity (ϕ), permeability (K), density (ρ), natural gamma ray, seismic velocity ...) of carbonate rocks is important in understanding the nature of reservoirs, both for groundwater aquifers as well as for hydrocarbon traps. Origin and distribution of porosity and permeability are important parameters controlling the transport phenomena within and through reservoir rocks (Lucia, 1995). Diagenetic and tectonic features such as cementation and fracturing in carbonates largely control essential reservoir porosity and permeability and can create significant reservoir anisotropy (Nelson, 1985). Primary permeability in limestones is often low but the effects of carbonate dissolution (Worthington and Smart, 2004) enhance secondary permeability.

Middle Miocene shallow-water carbonates are widespread in the Paratethys Sea and build reservoirs and potential exploration targets in several basins such as the Vienna Basin, the Styrian basin and the Pannonian Basin (e.g. Sauer et al., 1992; Strauss et al., 2006; Schreilechner and Sachsenhofer, 2007). In Austria, Slovakia and Hungary, these shallow-water carbonates (Badenian of the regional Central Paratethys stratigraphy, e.g. Hohenegger et al., 2014) are summarized under the term Leitha limestones (Papp et al., 1978; Riegl and Piller, 2000). Leitha limestones is a lithostratigraphic unit composed mainly of coralline algae (Riegl and Piller, 2000), deposited in middle to late Badenian (corresponds to Langhian to early Serravalian) and widespread in the Central Paratethys (Reuter et al., 2012; Wiedl et al., 2012, 2013, 2014; Hohenegger and Wagreich, 2012; Hohenegger et al., 2014). Physical properties of Leitha limestones were studied in detail as these rocks were widely used as a building stone in historical buildings in Austria, especially in Vienna (Rohatsch, 2005, 2007, 2012; Bednarik et al., 2014). However, data on porosities and permeabilities are considerably sparse (Sauer et al., 1992).

In this paper six localities were selected for petrophysical investigations, namely Mannersdorf, Wöllersdorf, Bruckneudorf, Kummer, Rosenberg (Austria) and Fertőrákos (Hungary). The compilation of petrophysical parameters (e.g. porosity, permeability, density, P-wave velocity and magnetic susceptibility) were used to deduce the evolution of porosity as a function of depositional and diagenetic processes.

2. Geological settings

The studied Leitha limestones lie within three Central Paratethys basins (Fig. 1). The Vienna Basin includes Mannersdorf quarry, Wöllersdorf abandoned quarry and borehole Bruckneudorf, the Eisenstadt-Sopron Basin includes Kummer and Fertőrákos abandoned quarries and the Styrian Basin includes Rosenberg quarry. Vienna Basin is a rhomb shape structure formed as a pull-apart basin along the junction of the Eastern Alps and Western Carpathians (Royden, 1985; Decker and Peresson, 1996; Hölzel et al., 2010). Subsidence continued with

increasing rates during Badenian where islands and shoals represented shallow carbonate platform with extensive Leitha limestones deposition during Middle Miocene (Riegl and Piller, 2000; Schmid et al., 2001; Wagreich and Schmid, 2002; Harzhauser and Piller, 2010; Lee and Wagreich, 2017). The Vienna Basin is the first and largest hydrocarbon provider in Austria and shows various hydrocarbon fields (3500 wells) such as Neogene basin-fill sandstones, allochthonous Upper Triassic dolomites of the Calcareous Alpine Zone, Flysch sandstones and deeper to autochthonous sub-thrust reservoir of Jurassic age (Hamilton et al., 1999).

The Eisenstadt-Sopron Basin and the Styrian Basin are located at the westernmost part of the Pannonian Basin and represent extensional basins in the Early Miocene during the final stage of Alpine orogeny (Ebner and Sachsenhofer, 1995; Sachsenhofer, 1996). The Styrian Basin was divided into the Eastern and Western Styrian subbasins by the N-S trending Middle Styrian Hills. Subsidence and eruptive volcanism due to extensional tectonics took place during the Badenian (Kröll, 1988; Seghedi and Downes, 2011).

3. Materials and methods

The rock samples were collected from limestone outcrops from five quarries (Mannersdorf, Wöllersdorf, Kummer, Fertőrákos and Rosenberg) and a cored groundwater well at Bruckneudorf. 67 thin sections of 5 × 5 cm dimensions were prepared for facies, pore space and cement investigations. Thin sections were examined using Leica microscope (Leica DM2700P) and photomicrographs were taken with Leica camera (Leica MC170 HD) connected to computer software (LAS v4.4.0). Scanning Electron FEI Inspect S- SEM at the University of Vienna, Faculty of Geosciences, Geography and Astronomy, was used for SEM photographs. The apparatus is equipped with secondary electron detectors for operation at high and low vacuum, as well as a back scattered electron detector and an energy dispersive X-ray detection unit (EDAX Apollo XV). Plugs of 3.8 cm diameter and up to 7 cm length were drilled using Optimum drilling device (OPTi drill B 26Pro) at University of Vienna. The edges of the plugs were cut at right angles using Remet TR60 device where the plug is horizontally fixed and a vertical moving saw cuts the edges of the plugs. Due to the soft nature of some of the limestones, the preparation process and the availability of material from outcrops the number of the obtained plugs was restricted to 38 representing various facies types of Leitha limestones.

Water porosity was measured by immersing the samples in water (triple weight method) until a constant mass is reached. The mass was measured in time interval of 1, 2, 3 and 6 days and the last measurements after a constant weight was obtained. Gas porosity and permeability were measured using Coreval 700 (Vinci Technologies) in the petrophysics laboratory at University of Vienna. The device measures the grain and bulk densities (ρ_g and ρ_b , respectively), pore and bulk volumes (V_p and V_b , respectively), gas porosity (ϕ) and gas permeability (K). These parameters were measured under different values of confining pressure (i.e. 400, 600 and 1000 psi). In low porous and low permeable samples only one measurement was done under 400 psi confining pressure. P-wave velocity and Magnetic Susceptibility were measured on the plugs using the Geotech bare-rock p-wave transducers and the Bartington MS2E magnetic susceptibility operated in a manual modes using the Geotech Multi-Sensor-Core-Logger (MSCL) System at the University of Innsbruck, Austria.

4. Results

4.1. Facies types

Leitha limestones exhibit different petrographic features among the studied localities based mainly on primary depositional features. Three lithologic facies occur in the studied samples from Mannersdorf, Wöllersdorf and Bruckneudorf limestones: 1) soft rudstone with

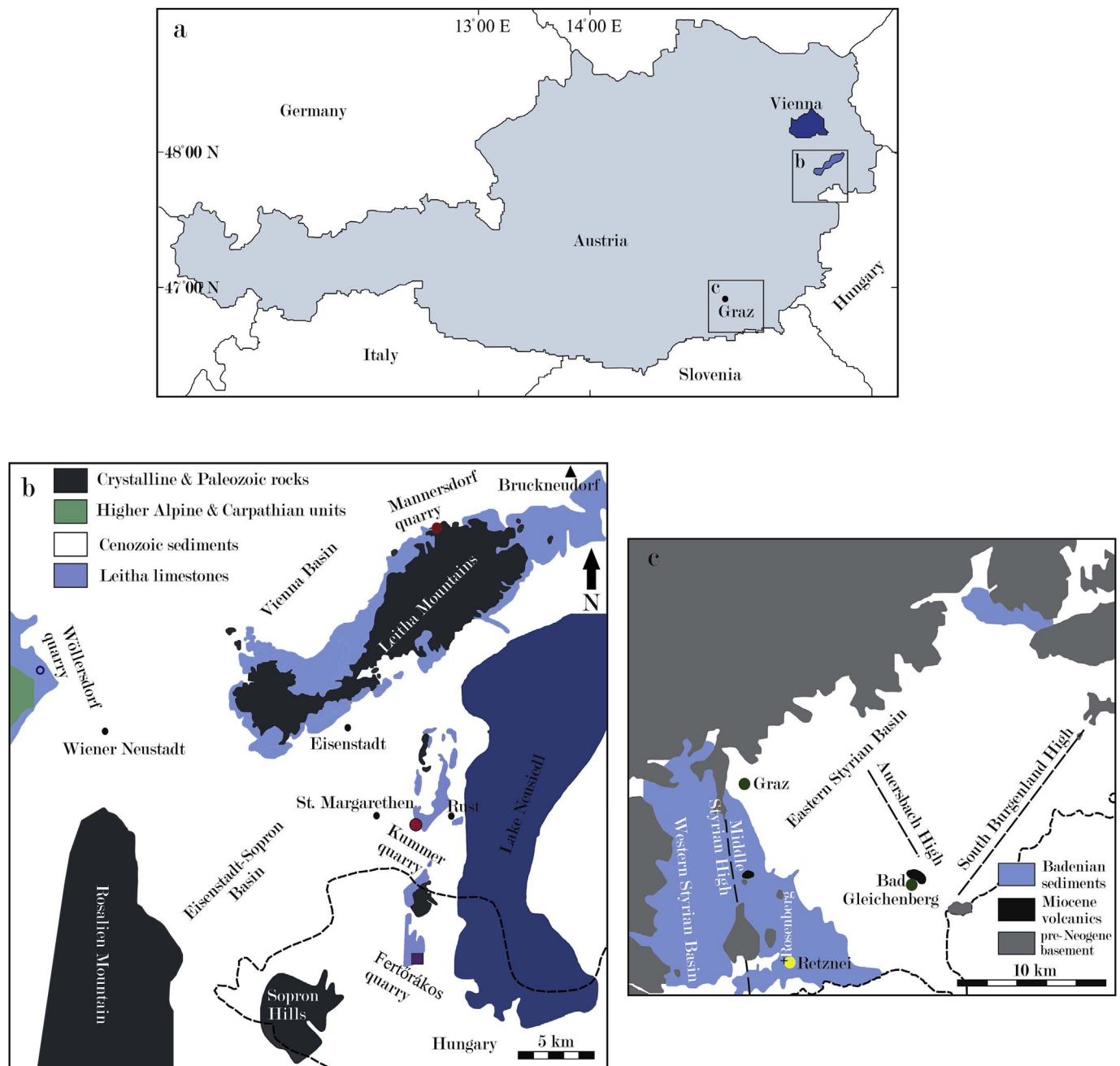


Fig. 1. a) Map of Austria with highlight on the studied basins, b) simplified geological map of the southern Vienna Basin and Eisenstadt-Sopron Basin with location of the studied areas (after Strauss et al., 2006; Schmid et al., 2001), c) simplified geological map of the Styrian Basin with location of Rosenberg quarry (after Reuter et al., 2012).

packstone matrix, which is pale yellow to white in colour and soft in hand specimens occur only in Mannersdorf samples (Fig. 2a); 2) poorly-sorted massive rudstone with packstone matrix, which is grey to beige in colour and hard in hand specimens (Fig. 2b) and 3) dolomitic facies represented by one sample from Wöllersdorf. Kummer limestones display two lithologic facies: 1) well sorted medium grained grainstone yellow to white coloured and soft in hand specimens (Fig. 2c); 2) poorly sorted coarse grained rudstone brown in colour and hard in hand specimens (Fig. 2d), these rocks suffered compaction, which resulted from pressure solution along the grain contact. In addition, Rosenberg limestones samples comprise lithologies of bluish grey wackestone-packstone to white packstone (Fig. 2e).

4.2. Pore space classification

Pore space in carbonates are principally classified into interparticle and vuggy pores, with the latter subdivided into separate and touching vuggy pores (Lucia, 1983, 1995). Mannersdorf limestones show various types of pore space including interparticle and vuggy pore space depending on the lithology. The soft rudstone samples are highly porous and the interparticle pores are connected to each other (Fig. 3a). The vuggy pores are of the separate type only and in some cases connected through interparticle pores. Different types are recorded such as moldic pores, in which fossil shells are totally dissolved (Fig. 3b). Intrafossil pore spaces are common and are up to 3 mm across. The massive rudstone samples are less porous and close-packed. Pores are not connected and show a high degree of cementation (Fig. 3c). Micropores exist between cement crystals and are visible only under SEM (Fig. 3d). In

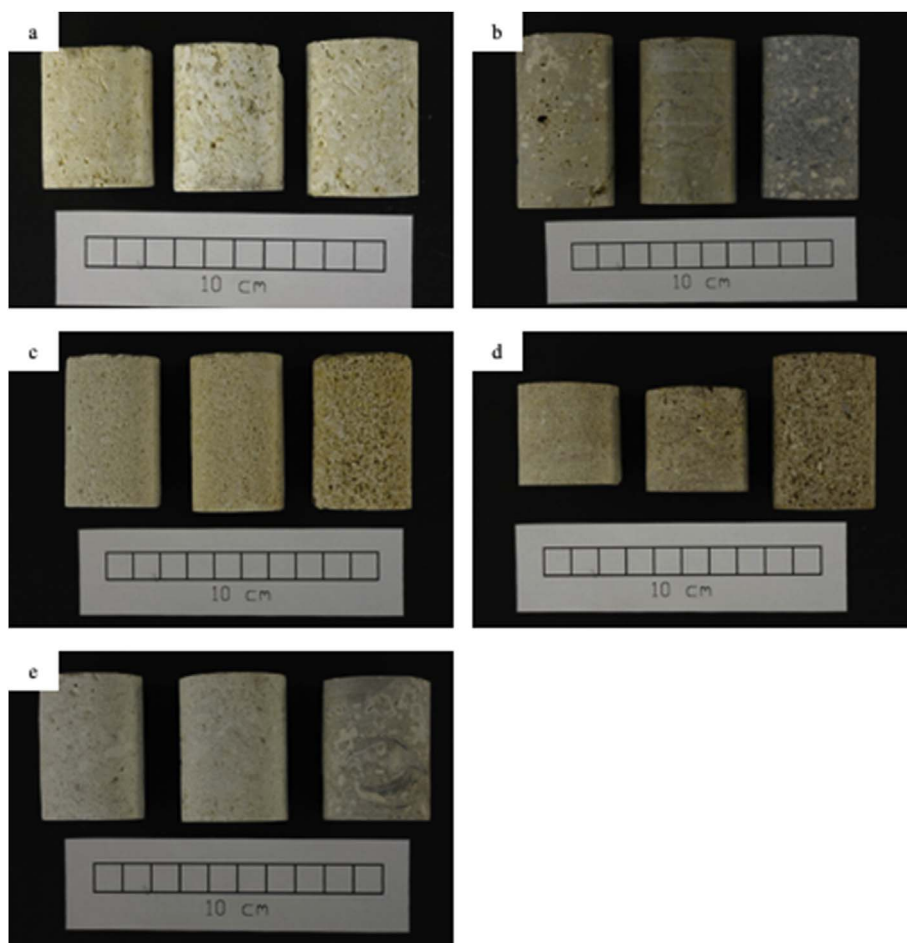


Fig. 2. Representative plugs of limestone lithofacies types; a) soft rudstone pale yellow to white in colour, Mannersdorf; b) grey to beige massive rudstone with karst (left), Mannersdorf, Wöllersdorf and Bruckneudorf; c) medium grained soft grainstone yellow to white in colour; d) massive coarse grained rudstone brown in colour, Kummer; e) Rosenberg limestones bluish grey wackestone-packstone to white packstone. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Wöllersdorf massive rudstone samples both interparticle and intrafossil pores exist and show variable degree of cementation.

Grainstone facies of Kummer and Fertőrákos limestones exhibit both

interparticle and vuggy pore space. In comparison with the former localities, the interparticle pore spaces are larger and well connected (Fig. 4 a). Also, vuggy pore spaces are widespread and include moldic

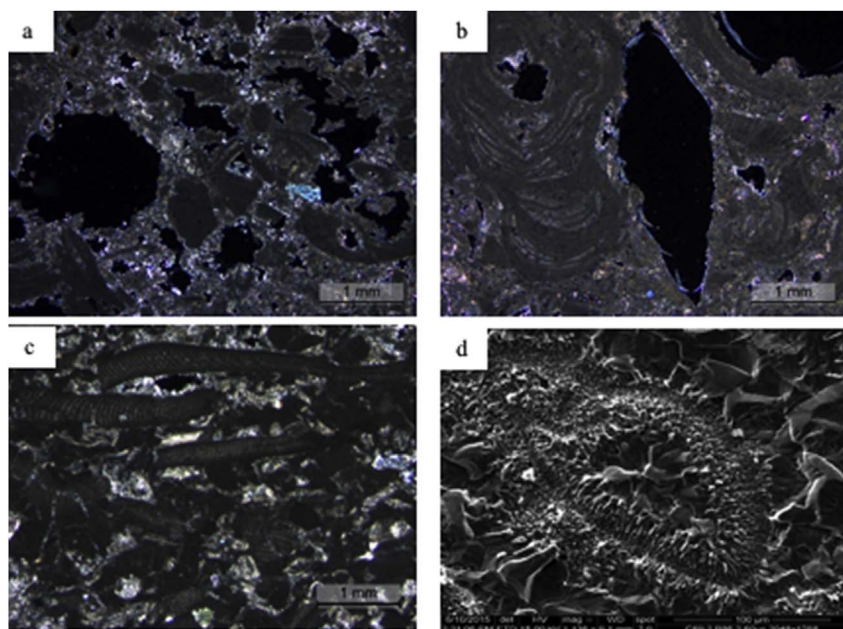


Fig. 3. a) Soft rudstone with large and well connected pore spaces, (crossed polars, XPL), M14/27; b) soft rudstone with moldic porosity where foraminifera shell is totally dissolved, XPL, M14/33; c) isolated interparticle pore space with growing cement in close-packed massive rudstone, XPL, M14/3; d) SEM image of massive rudstone with small intrafossil pore space in well cemented chamber, M14/9.

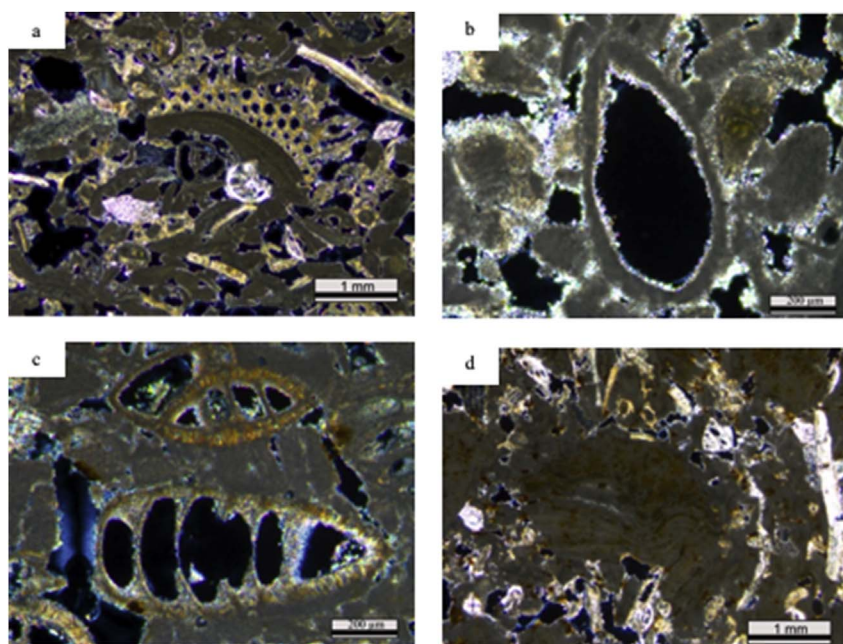


Fig. 4. a) Grainstone facies with connected interparticle pores and intrafossil pores, XPL, SK/1; b) moldic pores in a dissolved shell, XPL, F/3; c) intrafossil pores in foraminifer shells, XPL, SK/12; d) interparticle pores in massive rudstone facies and pressure solution in red algae (center), XPL, SK/17. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

pores (Fig. 4b) and intrafossil pores, which occur in all fossil types such as red algae, bryozoans (Fig. 4a) and foraminifers (Fig. 4c). The massive rudstone facies in Kummer limestones display less pore spaces in both size and type. The interparticle pores are small in size and fairly connected (Fig. 4d). The vuggy pores are subordinate and moldic pores are absent. Rosenberg limestones' both facies types are low porous and the pores are of the intergranular type and intrafossil type.

4.3. Cement types

Cement is a common porosity occluding material, which can form in a variety of environments. In soft rudstones from Mannersdorf the cement is less abundant and fills partially the interparticle pores such as

dog tooth cement type (Fig. 5a). The massive rudstone facies limestones are well cemented and sparry calcite cement is abundant (Fig. 5b); dog tooth cements also exist (Fig. 5c). These two types of cement occur also in Wöllersdorf limestones. In Kummer and Fertőrákos grainstones cementation took place along grain contacts and formed a thin layer of cement in the interparticle and intraparticle pores; also dog tooth cement occurs (Fig. 5d). Blocky sparite cement is rarely observed. The massive rudstones have similar cements besides pressure solution (Fig. 4d). In Rosenberg limestones blocky sparite cement predominates and fills the interparticle pores and intrafossil pores in bryozoans and red algae.

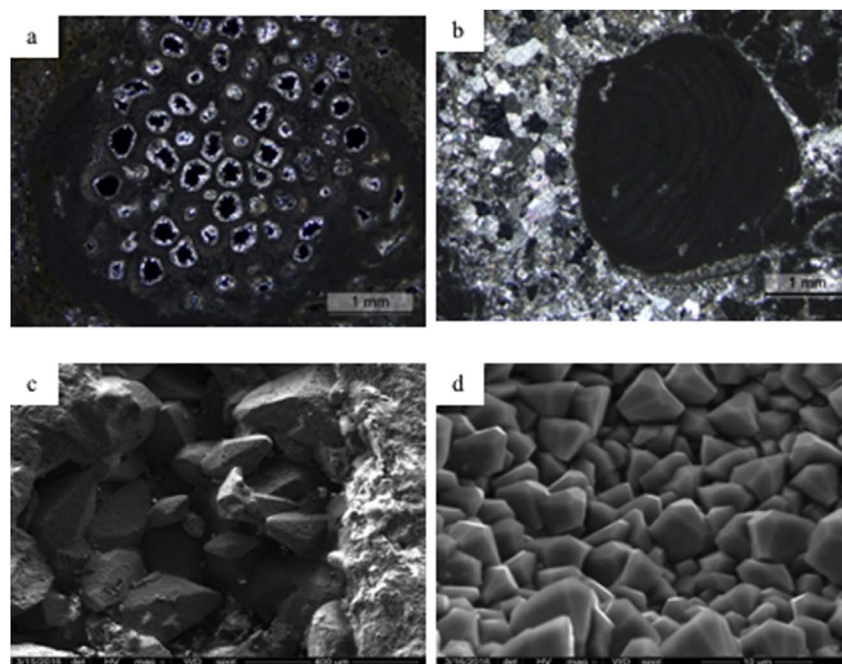


Fig. 5. a) Dog tooth cement in soft rudstone, XPL, M14/25; b) sparry calcite cement filling the primary depositional porosity, XPL, M14/10; c) SEM image of dog tooth cement in massive rudstone, M14/41; d) SEM image of dog tooth cement inside a foraminifer chamber, SK/18.

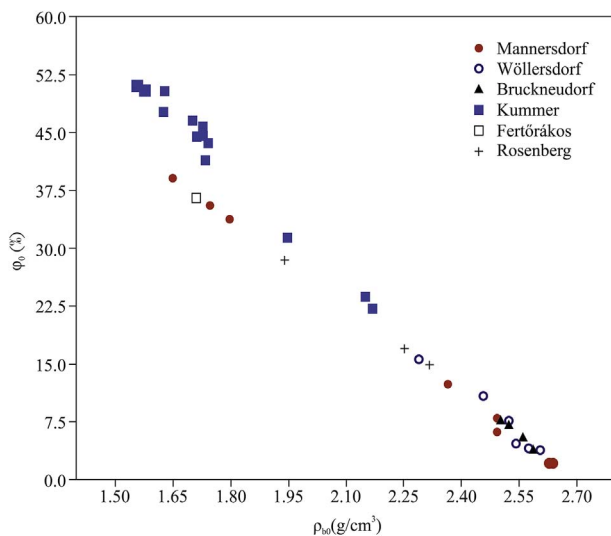


Fig. 6. Ambient bulk density vs. porosity at ambient pressure with a clear reverse relation.

4.4. Bulk and grain densities

The bulk density of the investigated limestones facies measured in dry samples is, as expected, reversely proportional to the porosity (Fig. 6). The soft rudstone facies from Mannersdorf limestones exhibit low bulk densities at ambient pressure (ρ_{b0}) from 1.62 to 1.8 g/cm³, while the massive rudstone facies have higher bulk densities from 2.36 to 2.64 g/cm³. Wöllersdorf massive rudstone samples have bulk densities similar to that of Mannersdorf from 2.29 to 2.58 g/cm³ and the dolomitic sample has a value of 2.46 g/cm³ (Table 1). Bruckneudorf massive rudstone samples have bulk densities from 2.5 to 2.59 g/cm³. The grainstone facies from Kummer and Fertőrákos have low bulk densities ranges from 1.56 to 1.74 g/cm³. The massive rudstone facies

from Kummer have bulk densities ranges from 1.95 to 2.17 g/cm³, Rosenberg facies show values ranging from 1.94 to 2.32 g/cm³ (Table 2). The bulk densities increase with increasing the confining pressure (Tables 3 and 4).

The grain density values in the soft rudstone facies is 2.71 g/cm³ and ranges from 2.66 to 2.73 g/cm³ in the massive rudstone facies in Mannersdorf and Wöllersdorf samples and the highest in the dolomitic facies with 2.75 g/cm³ (Table 1). Bruckneudorf, Kummer, Fertőrákos and Rosenberg facies have a narrower range of grain densities from 2.69 to 2.72 g/cm³ (Table 2).

4.5. Porosity

Soft rudstone samples from Mannersdorf have water porosity (ϕ_w) ranging from 24.1 to 29.4% and massive rudstones have values ranging from 2.1 to 3.6% (Mannersdorf) and 2.5–10% (Wöllersdorf) while the dolomitic sample from Wöllersdorf has a value of 6.3% (Table 1). The grainstone samples from Kummer and Fertőrákos have water porosities range from 22.9 to 32.5% and the massive rudstone samples range from 11.4 to 22.2%. In Rosenberg samples the wackestone has lower water porosity of 12% while the packstone has higher values ranging from 15.2 to 20.9% (Table 2).

Gas porosity under 400 psi confining pressure in Mannersdorf soft rudstone samples is higher than the massive rudstone samples. It ranges from 32.4 to 38.2% in the former samples and from 0.9 to 11% in the later samples (Table 1). The massive rudstone facies from Wöllersdorf have values ranging from 1.2 to 14.9% and the dolomitic sample has 10.8%. The grainstone samples from Kummer have gas porosities under 400 psi confining pressure range from 36 to 42.6% and in massive rudstone range from 20.2 to 28.7%. The grainstone sample from Fertőrákos has gas porosity of 30.4%. Similar to water porosity in Rosenberg the wackestone sample has 13.8% gas porosity, which is lower compared to the packstone samples which range from 16.4 to 20.9% (Table 2). The gas porosity decreases with increasing the confining pressure in all facies (Tables 1 and 2).

Table 1

The variation in ρ_g , ρ_{b0} , ϕ_w , ϕ and K with confining pressure in limestones from Mannersdorf, Wöllersdorf and Bruckneudorf.

Facies type	Sample	ρ_g (g/cm ³)	ρ_{b0} (g/cm ³)	ϕ_w (%)	Confining pressure (psi)					
					400		600		1000	
					Φ (%)	K [N ₂] (mD)	Φ (%)	K [N ₂] (mD)	Φ (%)	K [N ₂] (mD)
Massive rudstone	M14/21	2.69	2.63	2.1	0.9	0.005	–	–	–	–
	M14/42	2.66	2.49	3.6	4.5	0.007	–	–	–	–
	M14/41	2.70	2.36	3.6	11	0.05	–	–	–	–
Soft rudstone	M14/34	2.71	1.67	27.5	35.6	108.7	–	–	–	–
	M14/27	2.71	1.80	24.1	32.4	1353	32.1	1329	31.8	1243
	M14/26	2.71	1.75	25.7	34.2	59.7	34.1	58.04	–	–
	M14/24	2.71	1.62	29.4	38.2	124.3	–	–	–	–
	M14/25	2.71	1.65	28.4	37.4	107.6	37.1	105.3	–	–
Massive rudstone	M14/5	2.70	2.49	2.7	6.9	1.21	6.6	1.12	–	–
	M14/12	2.69	2.64	2.1	2.2	0.033	1.8	0.001	–	–
	WÖ/1	2.68	2.58	3.1	1.2	0.007	0.99	0.003	–	–
	WÖ/2	2.71	2.29	10	14.9	68.5	14.4	68.3	14.3	67.8
	WÖ/3	2.67	2.54	2.5	3.7	0.004	–	–	–	–
	WÖ/4	2.73	2.52	4.0	6.7	0.02	–	–	–	–
	WÖ/6	2.70	2.60	3.1	3.2	0.04	–	–	–	–
	BD/13 m	2.71	2.50	–	4.9	0.006	–	–	–	–
	BD/29.9 m	2.72	2.52	–	5.7	0.009	–	–	–	–
	BD/37.15 m	2.71	2.56	–	3.4	0.004	–	–	–	–
	BD/37.4 m	2.69	2.59	–	2.7	0.004	–	–	–	–
Dolomitic	WÖ/5	2.75	2.46	6.3	10.8	0.07	10.7	0.06	10.6	0.06

– Not measured.

Table 2The variation in ρ_g , ρ_{bo} , ϕ and K with confining pressure in limestones from Kummer, Fertőrákos and Rosenberg.

Facies type	Sample	ρ_g (g/cm ³)	ρ_{bo} (g/cm ³)	ϕ_w (%)	Confining pressure (psi)					
					400		600		1000	
					Φ (%)	K [N ₂] (mD)	Φ (%)	K [N ₂] (mD)	Φ (%)	K [N ₂] (mD)
Well sorted grainstone	SK/1	2.71	1.70	28.3	37.6	7539	37.4	7472	37.2	7297
	SK/4	2.72	1.63	29.2	40.1	12657	40	12609	39.7	12564
	SK/8	2.70	1.73	25.2	36.6	1569	36.5	1563	36.2	1550
	SK/9	2.72	1.73	28.3	36.2	31238	35.9	30584	35.5	29409
	SK/10	2.70	1.74	27.1	36	10137	35.9	10049	35.7	9935
	SK/14	2.71	1.56	32.5	42.6	3179	42.3	3104	41.5	2873
Poorly sorted rudstone	SK/16	2.71	2.15	13.3	21.2	0.56	21.1	0.54	20.9	0.49
	SK/17	2.71	2.17	11.4	20.2	52.3	20.1	51.9	20	51.6
	SK/18	2.71	1.95	22.2	28.7	7938	28.6	7903	28.5	7906
Well sorted grainstone	SK/20	2.72	1.73	28.5	36.5	15495	36.4	15283	36.2	15164
	SK/23	2.71	1.71	30.7	38.9	26178	38.6	25687	38.1	24843
	SK/24	2.70	1.62	30.6	39.9	1367	39.8	1352	39.5	1327
	SK/27	2.71	1.58	30.9	42.5	6207	42.3	6180	41.9	6000
	F/3	2.69	1.71	22.9	30.4	2891	30.4	2878	30.2	2853
Packstone-wackestone	RR/4	2.71	1.94	20.9	22.9	1.83	22.8	1.76	22.6	1.68
	RR/5	2.71	2.25	15.2	16.4	0.91	16.3	0.85	16.2	0.8
	RR/7	2.72	2.32	12	13.8	0.16	13.7	0.16	13.4	0.15

- Not measured.

Table 3The variation in V_p , V_b and ρ_b with confining pressure in limestones from Mannersdorf, Wöllersdorf and Bruckneudorf.

Facies type	Sample	Confining pressure (psi)								
		400			600			1000		
		V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)	V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)	V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)
Massive rudstone	M14/21	0.63	70.7	2.66	–	–	–	–	–	–
	M14/42	2.66	59.7	2.54	–	–	–	–	–	–
	M14/41	6.35	57.5	2.40	–	–	–	–	–	–
Soft rudstone	M14/34	19.3	54.3	1.75	–	–	–	–	–	–
	M14/27	20.6	63.4	1.83	20.3	63.1	1.84	19.9	62.8	1.85
	M14/26	19.9	58.1	1.78	19.8	58	1.79	–	–	–
	M14/24	18.7	49.1	1.68	–	–	–	–	–	–
	M14/25	21.9	58.6	1.70	21.7	58.3	1.70	–	–	–
Massive rudstone	M14/5	2.77	40.5	2.52	2.66	40.3	2.53	–	–	–
	M14/12	0.62	28.7	2.63	0.51	28.6	2.64	–	–	–
	WÖ/1	0.78	65.0	2.65	0.64	64.9	2.66	–	–	–
	WÖ/2	9.09	60.9	2.31	8.74	60.5	2.32	8.62	60.4	2.32
	WÖ/3	1.71	46.1	2.57	–	–	–	–	–	–
	WÖ/4	5.01	74.7	2.55	–	–	–	–	–	–
	WÖ/6	2.22	69.5	2.62	–	–	–	–	–	–
	BD/13 m	3.23	66.0	2.58	–	–	–	–	–	–
	BD/29.9 m	3.11	54.5	2.56	–	–	–	–	–	–
	BD/37.15 m	2.33	69.8	2.62	–	–	–	–	–	–
	BD/37.4 m	1.35	50.5	2.62	–	–	–	–	–	–
Dolomitic	WÖ/5	7.03	65	2.46	6.97	64.9	2.46	6.87	64.8	2.46

- Not measured.

4.6. Gas porosity and confining pressure

The measured samples under different confining pressures (i.e. 400, 600 and 1000 psi) exhibit, as expected, a decrease in porosity and permeability. For instance, the porosity decrease percentage (e.g. 400 to 600 psi confining pressure) in most samples is less than 1% in all facies except in the massive rudstone samples M14/5, M14/12, WÖ/1 and WÖ/2 where the difference varies from 4.3 to 18.2%.

4.7. Permeability

The gas permeability in Mannersdorf soft rudstone samples ranges from 59.7 to 1353 mD and is generally lower in massive rudstone samples (0.01–1.21 mD in Mannersdorf, Wöllersdorf and Bruckneudorf) except one sample which has a higher permeability of 68.5 mD while the dolomitic sample has also a distinct low permeability value of 0.07 mD (Table 1). In Kummer and Fertőrákos grainstone samples have very

Table 4The variation in V_p , V_b and ρ_b with confining pressure in limestones from Kummer, Fertőrákos and Rosenberg.

Facies type	Sample	Confining pressure (psi)								
		400			600			1000		
		V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)	V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)	V_p (cm ³)	V_b (cm ³)	ρ_b (g/cm ³)
Well sorted grainstone	SK/1	25.7	68.5	1.69	25.5	68.3	1.69	25.3	68	1.70
	SK/4	21.9	54.6	1.63	21.8	54.5	1.63	21.5	54.2	1.64
	SK/8	19.8	54.1	1.71	19.7	53.9	1.72	19.4	53.7	1.73
	SK/9	22.3	61.6	1.73	22	61.3	1.74	21.6	60.9	1.75
	SK/10	16.5	45.7	1.73	16.4	45.6	1.73	16.2	45.5	1.74
	SK/14	12.8	30.1	1.56	12.7	30	1.56	12.3	29.5	1.59
Poorly sorted rudstone	SK/16	10.2	48.3	2.13	10.2	48.2	2.14	10	48.1	2.14
	SK/17	8.3	41	2.17	8.2	40.9	2.17	8.2	40.9	2.17
	SK/18	17.4	60.6	1.93	17.3	60.5	1.94	17.2	60.4	1.94
Well sorted grainstone	SK/20	27.6	75.8	1.72	27.5	75.6	1.73	27.3	75.4	1.73
	SK/23	25.5	65.6	1.66	25.2	65.3	1.67	24.7	64.8	1.68
	SK/24	21.3	53.3	1.62	21.2	53.2	1.63	21	53	1.63
	SK/27	23.2	54.6	1.56	23.1	54.5	1.56	22.6	54.1	1.58
	F/3	17.4	57.1	1.87	17.3	57.1	1.87	17.2	56.9	1.88
Packstone-wackestone	RR/4	12.2	53.1	2.09	12.1	53	2.09	12	52.9	2.1
	RR/5	9.4	57.2	2.27	9.3	57.1	2.27	9.3	57.1	2.27
	RR/7	9.4	68.4	2.35	9.4	68.3	2.35	9.1	68.1	2.36

- Not measured.

high gas permeabilities ranging from 1367 to 31238 mD. The massive rudstone samples have lower permeabilities of 0.56, 52.3 and 7938 mD for the three samples under 400 psi confining pressure. From Rosenberg quarry, both facies have low permeabilities ranging from 0.16 to 1.83 mD (Table 2). The gas permeability reduces as the confining pressure increases (Tables 1 and 2).

4.8. P-wave velocity (P-wave)

The P-wave velocity values are highly variable in the sampled limestones facies. A clear proportional correlation with bulk densities ($r = 0.93$, Fig. 7) and reverse correlation with porosities at ambient pressure ($r = -0.92$) is evident in all studied samples. Mannersdorf soft rudstone samples have low P-wave velocities (in average 2636 m/s), while massive rudstone samples have higher P-wave velocities (in average 6103 m/s). Massive rudstone samples from Wöllersdorf have an

average P-wave velocity of 5531 m/s and the dolomitic sample has a value of 4753 m/s. Bruckneudorf massive rudstone samples record also high P-wave velocity (in average 6020 m/s). Grainstone samples from Kummer and Fertőrákos have an average P-wave velocity of 2706 m/s and the massive rudstone have higher velocities (in average 4163 m/s). Rosenberg samples have velocities of 2552 m/s for the packstone sample and higher velocities for the two wackestone samples (in average 3903 m/s) (Table 5).

4.9. Magnetic susceptibility

Magnetic susceptibility values of the investigated samples are negative in most samples ($n = 25$) and range from -0.002 to -1.252×10^{-5} SI units and positive in some samples ($n = 7$) and range from 0.065 to 1.325×10^{-5} SI units (Table 5).

The above-mentioned parameters (i.e. water porosity, gas porosities at 400 psi and ambient pressures, permeability at 400 psi, P-wave velocity and magnetic susceptibility) are plotted in a stratigraphic order at Mannersdorf and Kummer to evaluate petrophysics along reservoir profiles. More than 20 m thick highly porous and highly permeable limestones occur at Mannersdorf underlain and overlain by low porous and low permeable limestones (Fig. 8). At Kummer the situation is reversed, where the highly porous and extremely permeable limestones are interlayered by an interval of low porous and low permeable limestones (Fig. 9).

5. Discussion

All petrophysical parameters investigated in this study indicate various depositional and diagenetic criteria of Leitha limestones. In Mannersdorf, Wöllersdorf and Bruckneudorf the diagenetic alteration is highly significant. The soft rudstone facies in Mannersdorf with large pores, less cement, dissolution features and high porosity and permeability is equivalent to the interval with *Pinna* subfacies described by Wiedl et al. (2012) and was interpreted to be deposited in (< 10 m) moderately agitated shallow water within seagrass meadows. The petrographic and petrophysical features of these limestones point out to the existence of two main diagenetic environments represented by vadose

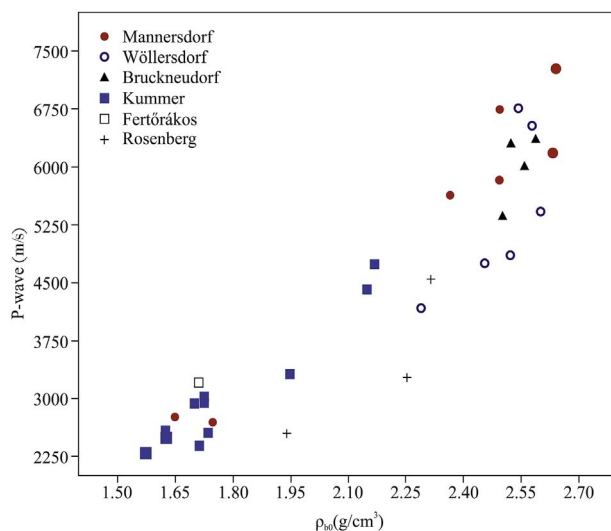


Fig. 7. P-wave velocity as a function of ambient bulk density with proportional correlation.

Table 5

P-wave velocity (m/s), gas porosity at ambient pressure (φ_0 (%)) and magnetic susceptibility (Mag Susc $\times 10^{-5}$ SI) of Leitha limestones.

Facies type	Sample	P-wave	φ_0 (%)	Mag Susc
Massive rudstone	M14/21	6197	2	0.065
	M14/42	5831	6.1	–
	M14/41	5639	12.4	–0.86
soft rudstone	M14/34	2798	45.3	–0.462
	M14/26	2695	35.5	–0.527
	M14/24	2297	47.6	–0.861
	M14/25	2753	39.1	–0.791
Massive rudstone	M14/5	6745	7.8	–1.252
	M14/12	7281*	1.9	–1.253
	WÖ/1	6541	4	–1.124
	WÖ/2	4172	15.6	–0.723
	WÖ/3	6761	4.6	–0.861
	WÖ/4	4858	7.5	–0.73
	WÖ/6	5424	3.7	–0.927
	BD/13 m	5354	7.7	–0.332
	BD/29.9 m	6320	7.1	–1.126
	BD/37.15 m	6026	5.4	0.465
	BD/37.4 m	6380	3.9	–0.762
dolomitic	WÖ/5	4753	10.8	–0.526
Soft grainstone	SK/1	2933	46.5	–0.33
	SK/4	2526	50.3	–0.327
	SK/8	2944	45.8	–0.002
	SK/9	2563	41.3	–0.065
Massive rudstone	SK/16	4429	23.6	1.325
	SK/17	4737	22.3	1.125
	SK/18	3324	31.4	1.056
Soft grainstone	SK/20	3023	44.6	–0.066
	SK/23	2391	44.4	0.068
	SK/24	2460	47.7	–0.333
	SK/27	2293	50.4	–
	H/3	3219	36.4	0.332
Packstone	RR/4	2552	28.4	–1.06
Wackestone	RR/5	3262	17	–1.06
	RR/7	4544	11.4	–1.06

– Not measured *The sample is short (2.5 cm) and excluded in discussion.

and phreatic meteoric environments. The vadose zone is confirmed by karst features, cementation confined to grain contact and preservation of depositional interparticle pores. Processes in the phreatic meteoric zone with less saturated water in respect to carbonate minerals enhanced porosity of the soft rudstone facies with dissolution features such as moldic porosity and enlarged interparticle pores.

The massive rudstone from these localities is related to the mollusc subfacies (Wiedl et al., 2012) which dominates at Mannersdorf and represents a shallow transition zone between shallower to deeper waters of the platform. The facies was affected by diagenesis of the phreatic meteoric zone with saturated water in respect to carbonate minerals. This resulted in progressive calcite cementation and reduced primary porosity to nil. Later dissolution and karstification of these limestones are a consequence of subaerial exposure (Tucker, 1993; Harris et al., 1999; Schlager, 2005; Moore and Wade, 2013).

Kummer and Fertőrákos Leitha limestones paleoenvironments were interpreted as small depressions surrounded by carbonate sand shoals (Schmid et al., 2001). Well sorted grainstones have high porosity and permeability and display many phases of cementation and dissolution. Firstly, a thin layer of marine cement was precipitated along the grain boundaries, followed by fresh water dissolution of metastable carbonate minerals resulting in moldic and intraparticle pore production. Phreatic blocky sparite cement is very rare in this facies.

The sedimentation was affected by synsedimentary faulting, i.e. the Eisenstadt fault (ESE-WNW extensional deformation, Fodor, 1995; Decker, 1996) where deformation bands, compaction and deformation of the massive rudstone facies were formed. Although this rock type has excellent reservoir capacity it should be taken into account that later tectonic features such as deformation bands are recorded in the grainstone. Deformation bands reduced initially high porosity locally along planes down to 2% (Rath et al., 2011; Lommatzsch et al., 2015a,b) and reduces the quality of the reservoir rocks in compartmentalizing reservoirs. The massive rudstone facies with lower porosity and permeability is a consequence of compaction which resulted in pressure solution and occlusion of the pores. An important parameter in permeability measurements is the flow direction where horizontally oriented samples are extremely permeable compared to the vertical permeability ($K = 0.56$ mD, SK/16). Similar diagenetic features of Leitha limestones were proposed for the Eisenstadt-Sopron basin (Dullo, 1983; Boldizsár et al., 1991).

Leitha limestones of the Styrian Basin, i.e. from Rosenberg quarry, were deposited during sea level rise (TB 2.4, Haq et al., 1987; Strauss et al., 2006) and influenced by both terrigenous siliciclastic input and volcanoclastic sedimentation (Friebe, 1990; Reuter et al., 2012). They are characterized by primary cementation, which blocked the interparticle pores, followed by slight leaching and partial dissolution of shells and intensive phreatic meteoric blocky cementation. These criteria point out to the main role of primary depositional imprint on Leitha limestones petrophysics compared to the diagenetic influence. The effect of diagenesis on these limestones is not such important in comparison to the other basins (Bojar et al., 2004; Ali and Wagreich, 2017).

The inverse correlation of P-wave and porosity indicates the great effect of the latter on the P-wave velocity, which is a function of depositional and diagenetic processes. Moreover, similar values of magnetic susceptibility of the limestones under investigation point to the negligible primary effect of mineral composition on P-wave velocity. The soft and porous limestones from the studied localities have low P-wave velocities while the massive and less porous limestones have higher velocities. In a reverse manner velocities are proportionally correlated to bulk densities. Limestones samples with the same porosities sometimes display different velocities, which reveal the role of pore types on the velocities. Limestones with dominant interparticle pores have lower P-wave velocities compared to limestones with prevailing moldic pores. The studied limestones are composed of calcite, which diminish the effect of mineralogy on P-wave velocities. In carbonate rocks, P-wave velocities are a factor of depositional mineralogy and diagenesis that modifies porosity and pore types (Anselmetti and Eberli, 1993, 2001; Verwer et al., 2008; Weger et al., 2009).

The porosity evolution of Leitha limestones exhibit different stages in the studied localities (Fig. 10). Mannersdorf soft rudstone samples with porosity up to 47.6% possess depositional interparticle porosities and less cementation and dissolution producing vuggy pores. As a consequence of the coarse grains of the facies, the main diagenetic phase that affected the limestones is dissolution of the shells. All these criteria result in reduced (up to 3000 m/s slower) p-wave velocities (Fig. 10a). On the other hand, the massive rudstone from Mannersdorf, Wöllersdorf and Bruckneudorf show higher P-wave velocities from 4858 to 6761 m/s due to blocky cementation, which blocks most of the pores and reduces the porosity to less than 2% in some samples (Fig. 10a and b). It is apparent that dissolution resulting in the formation of karst and moldic pores played a role in these types of limestones where samples with higher porosities, for example M14/5 ($\varphi_0 = 7.8\%$), have higher P-wave velocities (6745 m/s) compared to lower porosities, e.g. M14/42 ($\varphi_0 = 6.1\%$), which has lower P-wave velocities (5639 m/s). In Kummer limestones the porosity evolution path is affected by both depositional and diagenetic processes, where the well sorted grainstones deposited in high energy environment show early cementation and effect of dissolution which produces more moldic pores with higher

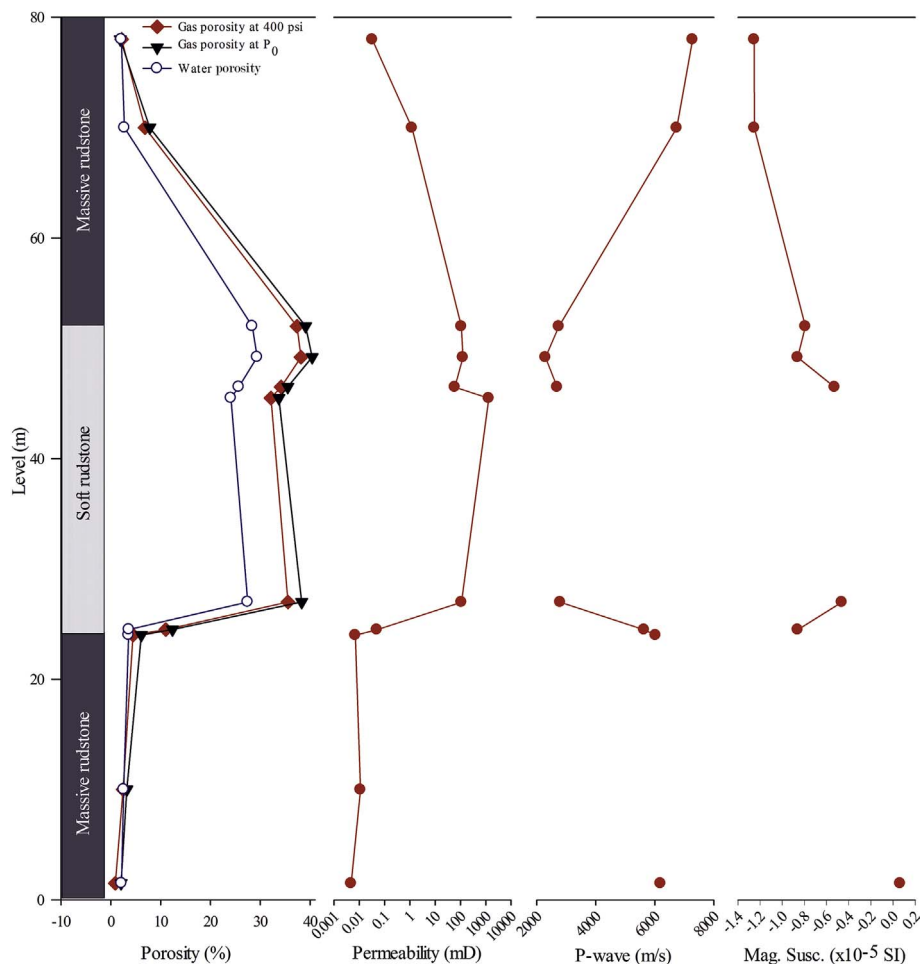


Fig. 8. Different petrophysical parameters of Leitha limestones at Mannsrdorf in a stratigraphic order, Mag. Susc. = magnetic susceptibility.

P-wave velocities compared to the same facies type samples with same porosities. In contrast, the poorly sorted massive rudstone was deposited in a low energy environment and displays different diagenetic features such as pressure solutions and higher degree of cementation that resulted in lower porosities and higher P-wave velocities (Fig. 10c). Rosenberg limestones display progressive cementation, which influenced the porosity and velocity with decreasing the former and increasing the later (Fig. 10a).

Combining all localities together to get a general diagenetic history of Leitha limestones (Fig. 10d), five principal successive, but overlapping stages of diagenesis can be deduced.

- 1) Early cementation of the sediments and formation of lithified limestones with high porosity under the influence of marine pore waters (Fig. 11). The age of this phase is constrained by syndepositional to early diagenetic features of marine pore waters that are restricted to Badenian and partly Sarmatian age, so a syn-to early post-depositional age, i.e. a middle to late Badenian age, is most probable.
- 2) Dissolution of some shells and moldic pores production probably due to early subaerial exposure especially in the Eisenstadt-Sopron Basin, which have higher velocities than interparticle pores in limestones with similar porosities. This phase is attributed mainly to the Sarmatian stage with well-known early erosion and reworking of Badenian limestones.
- 3) Progressive cementation due to burial during late Badenian, Sarmatian and Pannonian of both interparticle and vuggy pores resulting in lower porosities (and thus producing rock with higher P-Wave velocities).

- 4) Blocky cementation due to deeper burial as the last stage of cementation during Pannonian-Pliocene-Pleistocene and the occlusion of most pore spaces (resulting in rocks that show the highest P-wave velocities). Cementation ages are hard to ascertain for this gradual process. Maximum burial for the sites was reached during the Pannonian (Wagreich and Schmid, 2002), terminated by late Pannonian inversion, but local uplift along faults may have begun earlier or with different timing up to local Pleistocene basin development (Lee and Wagreich, 2017).
- 5) Further dissolution and karstification of the well-cemented limestones since late Pannonian-Pliocene till recent times due to uplift, exhumation and surface exposure, including ground water flow. This latest stage again enhanced the porosity, yet the petrophysical parameters remain characterized by high P-wave velocities (Likely due to the fact that elastic properties of the bulk rock retain). Karst features in some of the areas, e.g. around Mannsrdorf are mainly dated from Pliocene to Pleistocene (Wessely, 2006).

Porosity- P-wave velocity paths summarize the diagenetic history of the carbonate rocks. Diagenetic processes highly affect the P-wave velocity in various ways. Cementation of interparticle pores leads to increase in the velocity of the compressional wave and, when the blocky cementation occludes nearly all pores, the impedance become maximum and higher values of velocities are measured. Dissolution and moldic porosity formation increase the P-wave velocity and explains the different values of velocities for the same porosity values (Anselmetti and Eberli, 1993, 2001).

In comparison to geochemistry of Leitha limestones (Ali and Wagreich, 2017) the degree of cementation is correlated to the loss of

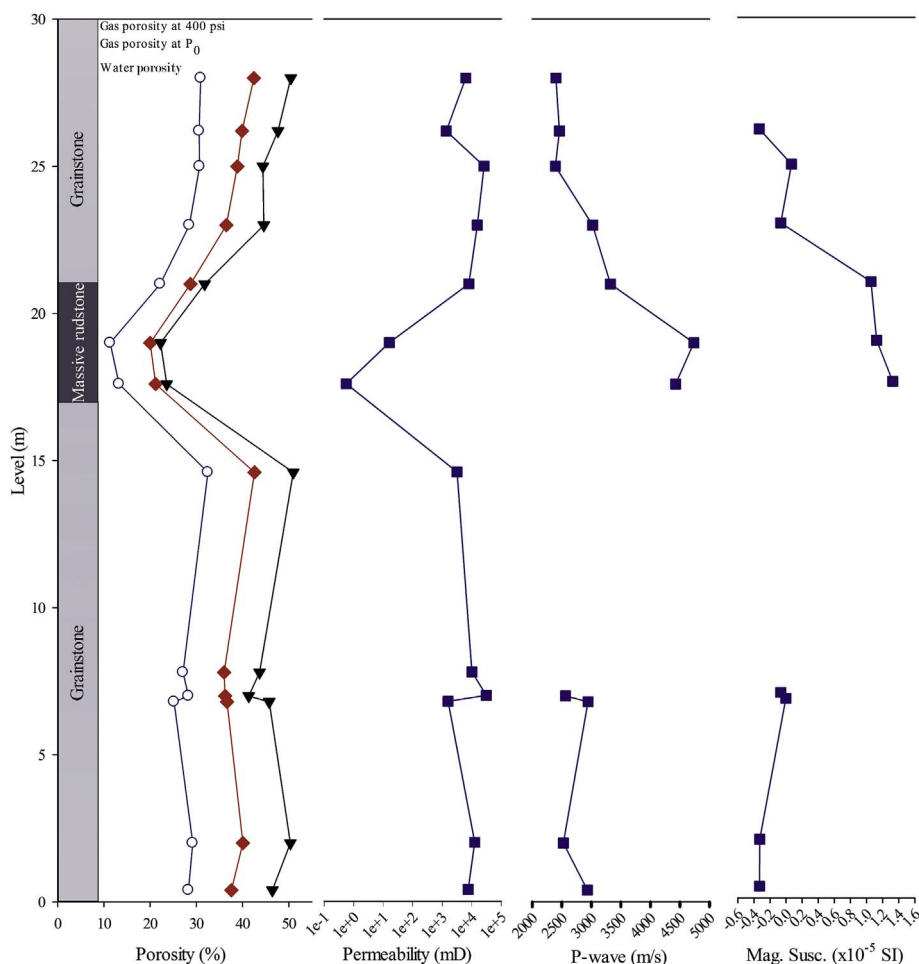


Fig. 9. Different petrophysical parameters of Leitha limestones at Kummer in a stratigraphic order.

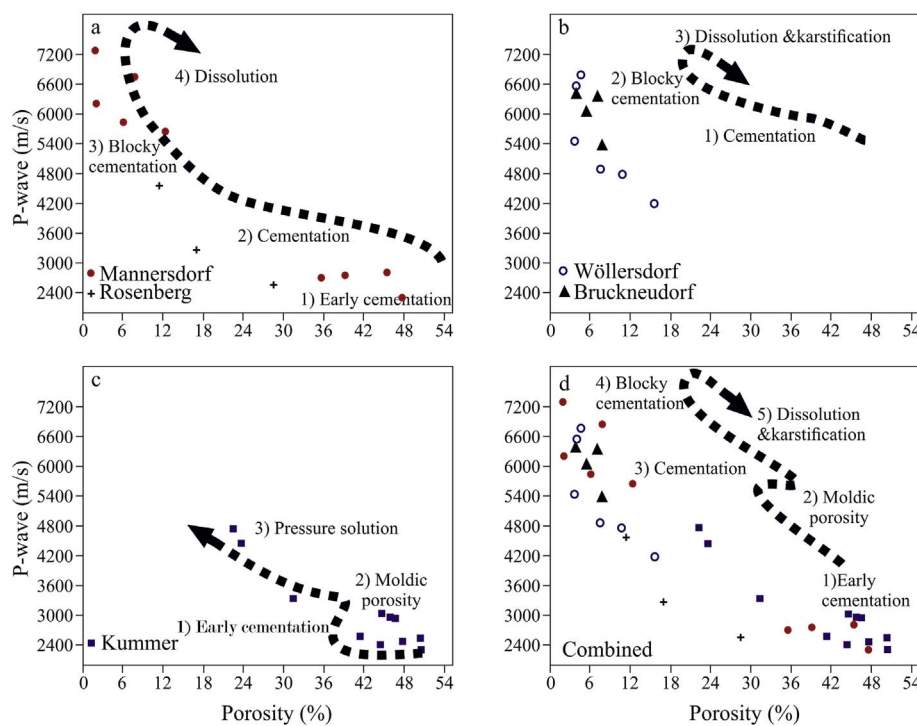


Fig. 10. Porosity-velocity path with different stages of diagenesis, a) Mannersdorf limestones affected by cementation and dissolution and Rosenberg limestones affected only by cementation; b) blocky cementation and later dissolution of Wöllersdorf and Bruckneudorf limestones; c) Kummer limestones displaying cementation, dissolution and pressure solution diagenesis and d) combined graph of all studied limestones show different diagenetic processes affected Leitha limestones (after Anselmetti and Eberli, 1993).

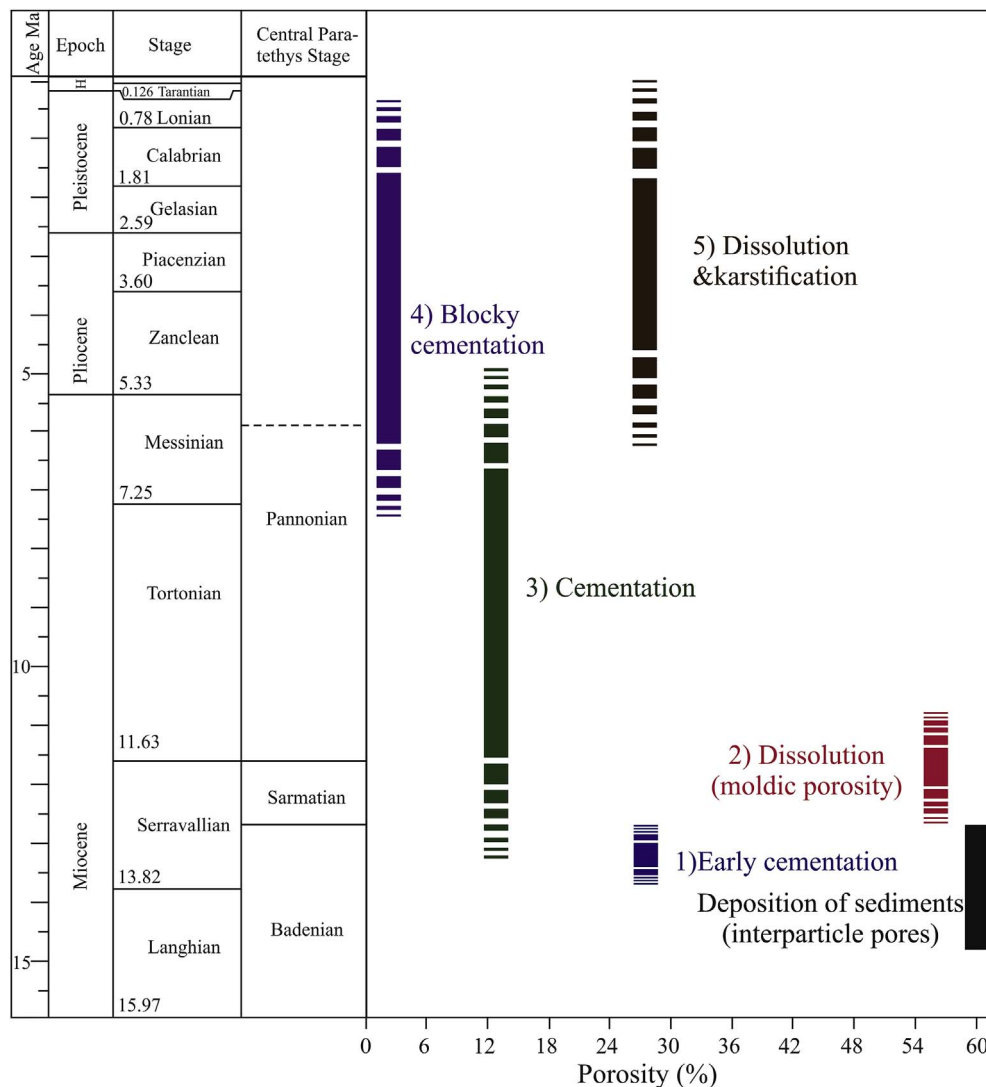


Fig. 11. The diagenetic evolution of Leitha limestones since deposition in the Middle Miocene Badenian stage till recent, there is no sharp contact of the diagenetic phases where many phases can take place simultaneously, the numeric ages after Gradstein et al. (2012) and the Central Paratethys stages are after Piller et al., 2007.

Sr. Mannersdorf and Wölldorf limestones exhibit a high degree of cementation and the lowest Sr content among the studied limestones. Kummer and Fertőrákos with lesser degree of cementation have higher Sr contents. Although, Rosenberg limestones have higher degree of cementation compared to Kummer and Fertőrákos they have higher Sr content, but these limestones were affected by the synsedimentary volcanic activity and thus Sr enrichment (Ali and Wagreich, 2017).

6. Conclusions

Leitha limestones from different basins are the result of locally varying depositional environments, and suffered different diagenesis. The Vienna Basin Leitha limestones occurrences have undergone several stages of diagenesis that started with early marine precipitation of marine cement. Following this early stage, dissolution took place followed by phreatic meteoric cementation and finally karstification due to subaerial exposure. These phases of diagenesis reduce the capacity of limestones as reservoirs for fluids, i.e. aquifers and hydrocarbon reservoirs.

In spite of this long-term diagenetic phases some facies were not affected by the phreatic meteoric environment and had the opportunity to retain the primary porosity which was enhanced due to the aggressive nature of the water through dissolution. The result of such

processes produced more than 20 m thick carbonate horizons with excellent reservoir potential. In the Eisenstadt-Soporon Basin Leitha limestones exhibit only early marine cementation and obvious impact of dissolution processes resulted in improving the secondary porosity of these rocks. Synsedimentary tectonics affected the limestones in this area and locally reduced the reservoir quality. The Styrian Basin locality was affected mainly by early marine cementation and to a lesser extent by blocky cementation and dissolution.

Leitha limestones cover a wide range of petrophysical properties in the outcrop due to depositional facies and diagenetic pathways. The depositional textures and limestones fabrics played the great role in the modification of the petrophysics of Leitha limestones. Results may help in deciphering potential and quality of carbonate reservoirs in the subground recognized by seismic data and predicted due to analogue studies.

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